



Haribhai V. Desai College of Arts, Science and Commerce, Pune. AUTONOMOUS

DEPARTMENT OF COMPUTER STUDIES

T.Y. B.Sc. CS - 50 PROJECTS for Field Project (FP)

Skills: PHP, Java, Python, Data Science, AWS, OS, DB

Goal: Full-stack, Cloud, Data, Deployable product. Use Python/Flask/Django/Java + MySQL + AWS Free Tier

Team Size: 3-4 students

Assessment in 3 Rounds: 1. Synopsis, 2. Mid Demo, 3. Final Demo

Note:

- Students are permitted to choose any one topic from the list below. Selection of a topic outside this list is permissible, subject to review and approval by the concerned authorities.
- Project guidelines will be provided by assigned FP faculty.

Domain-A: Web Development - PHP/Java - 12 Projects

1. College ERP Portal - PHP + MySQL + Roles
2. Online Learning Platform - Java Spring + Video upload
3. E-Commerce Website - PHP + Payment gateway mock
4. Job Portal - Java + Resume upload + Search
5. Hospital Management System - PHP + Appointment
6. Blog + CMS Platform - PHP + Admin panel
7. Online Examination System - Java + Timer + Auto grading
8. Real Estate Portal - PHP + Image upload
9. Alumni Portal - Java + Networking features
10. Event Management System - PHP + Ticketing
11. Restaurant Ordering System - Java + Cart
12. Social Media Mini Clone - PHP + Post, Like, Comment

Domain-B: Data Science + Python - 10 Projects

13. Early DR Detection Dashboard - Python + Streamlit + Pre-trained CNN
14. Student Dropout Prediction - Python + ML + Dashboard
15. Fake News Detector - NLP + Python + Flask
16. Sales Forecasting Dashboard - Python + Time Series + PowerBI
17. Chatbot for College Queries - Python + NLTK + Flask
18. Sentiment Analysis of Tweets - Python + ML
19. Crime Data Analysis - Python + Data Viz
20. Recommendation System - Python + Collaborative filtering
21. Fraud Detection in Transactions - Python + ML
22. Crop Yield Prediction - Python + ML + Data

Domain-C: Cloud + AWS + DevOps - 8 Projects

23. Cloud File Storage System - AWS S3 + PHP/Java
24. Serverless Student Attendance - AWS Lambda + API Gateway
25. Website Hosting on AWS - EC2 + LAMP stack deployment
26. CI/CD Pipeline Demo - GitHub + AWS CodePipeline
27. Database Backup on Cloud - AWS RDS + Automation
28. Chat App with AWS - EC2 + WebSockets
29. Image Recognition on Cloud - AWS Rekognition + Python
30. Cost Monitoring Dashboard - AWS Billing + Python

Domain- D: Advanced Applications - Java/Python - 10 Projects

31. Android App for College Notices - Java Android + Firebase
32. Library App with Barcode Scanner - Java Android
33. Banking System - Java + JDBC + Security
34. Hotel Management Desktop App - Java Swing + DB
35. Online Auction System - Python Django
36. Video Streaming Platform Mini - Python Flask
37. Inventory Management System - Java + Jasper Reports
38. Travel Booking System - Python Django + API
39. HR Management System - Java + Leave/Attendance
40. Fitness Tracker App - Python + Data Viz

Domain-E: Mini AI/IoT Simulation - 10 Projects

- 41. Spam Email Classifier - Rule based in C
- 42. Number Plate Detection Simulation - Image processing basics
- 43. Smart Home Device Simulator - C logic
- 44. Traffic Light Control Simulation - State machine in C
- 45. Chatbot - Rule based in C
- 46. Student Performance Predictor - Basic statistics in C
- 47. Password Strength Checker - String algorithms
- 48. Image Gallery Manager - File handling
- 49. Mini Search Engine - Indexing + DS
- 50. Voter Management System - DB + Validation
